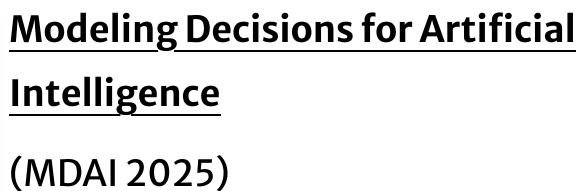


Measurable Closure of a Finitely-Additive Measure Space: An Analysis of Spaces Similar to Stone Spaces

| pp 3–14 | Cite this conference paper

Download book EPUB

[View saved research](#) >



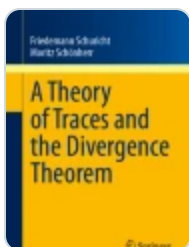
This website sets only cookies which are necessary for it to function. They are used to enable core functionality such as security, network management and accessibility. These cookies cannot be switched off in our systems. You may disable these by changing your browser settings, but this may affect how the website functions. Please view our privacy policy for further details on how we process your information. Dismiss

Abstract

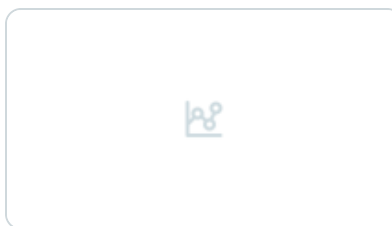
Finitely-additive set functions naturally arise in mathematical models; however, extending them to σ -additive measures is not always straightforward. For example, a finitely-additive measure defined on rational intervals may require space completion to achieve σ -additivity. A common approach involves the use of Stone spaces, which introduce a totally disconnected topology in order to ensure countable additivity. However, such methods often rely on non-constructive principles. In this paper, we propose a novel and intuitive framework for extending finitely-additive set functions by expanding the space, thereby clarifying the relationship between topological and measure-theoretic extensions.

 Access provided by COPPUL Room 219 Koerner Library. [Download conference paper PDF](#) 

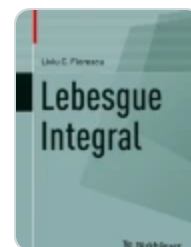
Similar content being viewed by others



Preliminaries About



Finitely additive



Signed Measures

This website sets only cookies which are necessary for it to function. They are used to enable core functionality such as security, network management and accessibility. These cookies cannot be switched off in our systems. You may disable these by changing your browser settings, but this may affect how the website functions. Please view our privacy policy for further details on how we process your information.

[Algebraic Topology](#)[Measure and Integration](#)[Proof Theory and Constructive Mathematics](#)[Sequences, Series, Summability](#)[Set Theory](#)[Functional Analysis](#)[Non-Additive Measure Theory and Fuzzy Integrals](#)

1 Introduction

Set functions have been used in various forms to describe properties associated with a space. As a basic example, the expected value of a fair die roll, assuming uniform distribution, is 3.5. In a more general situation where the corresponding probability measure is not uniform—such as when the die is irregularly shaped—the expected value expressed as an integral can take various values. Furthermore, when the space is an uncountable infinite set, this concept is abstracted and generalized as the Lebesgue integral with respect to a σ -additive measure. In cooperative game theory, allocations are often treated as finitely-additive set functions. However, when transitioning to a non-discrete setting and conducting analysis using measure theory, σ -additivity is essential for measure-theoretic analysis (see [1, 2] for example).

Since σ -additive measures are fundamental in modern analysis, we omit a detailed discussion here and refer readers to standard references [3]. When modeling an object of

This website sets only cookies which are necessary for it to function. They are used to enable core functionality such as security, network management and accessibility. These cookies cannot be switched off in our systems. You may disable these by changing your browser settings, but this may affect how the website functions. Please view our privacy policy for further details on how we process your information.

additive. The reason is that the value of the set function for a singleton set must be 0, and since every subset is countable, if countable additivity held, all values would have to be 0, contradicting the notion of interval length. In such a case, by completing the space, one can obtain the Lebesgue measure, which is a σ -additive measure. This study analyzes the structure of such extensions.

A well-known approach to achieving such extensions is the use of Stone spaces. The Stone space is a concept derived from the Stone Representation Theorem for Boolean algebras. By defining a totally disconnected topology on the space of ultrafilters or certain set functions, it becomes possible to extend finitely-additive set functions to be σ -additive. These theories are explained in [4] and [5]. Moreover, [4] employs this method to express the Choquet integral using a σ -additively extended Möbius transformation. The proofs of these results involve elegant arguments employing transcendental principles such as Zorn's Lemma. (These details are discussed in [5].) Since such extensions involve transcendent aspects, their fundamental nature can be challenging to grasp. In this study, we aim to provide an intuitively comprehensible method for extending finitely-additive set functions to be σ -additive by expanding the space. We anticipate that this perspective will elucidate the relationship between extensions of topological and measure-theoretic structures.

2 Cumulative Refinement Space

Let $(X, \mathcal{A}, \mu)$ be a finitely-additive measure space. that is, X is a set and $\mathcal{A}$ is an algebra over X , and μ is a nonnegative finitely-additive set function defined on $\mathcal{A}$. We assume that μ satisfies $\mu(X) < \infty$.

This website sets only cookies which are necessary for it to function. They are used to enable core functionality such as security, network management and accessibility. These cookies cannot be switched off in our systems. You may disable these by changing your browser settings, but this may affect how the website functions. Please view our privacy policy for further details on how we process your information.

sequences are used to capture limiting behavior and define elements of a larger, complete space.

$$\mathcal{R} = \mathcal{R}(\mathcal{A}) = \{ \mathbb{A} = \{A_n\}_{n \in \mathbb{N}} \mid A_n \in \mathcal{A}, A_{n+1} \subset A_n, \forall n \in \mathbb{N} \}.$$

We call the family $\mathcal{R}$ the space of cumulative refinements.

For $\mathbb{A} = \{A_n\}_{n \in \mathbb{N}} \in \mathcal{R}$, we define a natural extension $\underline{\mu}$ of μ as follows.

$$\underline{\mu}(\mathbb{A}) = \lim_{n \rightarrow \infty} \mu(A_n),$$

where $\mathbb{A} = \{A_n\}_{n \in \mathbb{N}} \in \mathcal{R}$.

Example 1

Let $X = [0, 1] \cap \mathbb{Q}$, where $\mathbb{Q}$ is the set of all rational numbers. Define

$$\mathcal{A} = \left\{ \bigcup_{j=1}^n [a_j, b_j) \mid n \in \mathbb{N}, a_j, b_j \in X, j \leq n, \text{ and these intervals are pairwise disjoint.} \right\}$$

and

This website sets only cookies which are necessary for it to function. They are used to enable core functionality such as security, network management and accessibility. These cookies cannot be switched off in our systems. You may disable these by changing your browser settings, but this may affect how the website functions. Please view our privacy policy for further details on how we process your information.

$$\mu_2([a, b)) = \begin{cases} 1 & \text{if } a \leq \frac{1}{2} < b \\ 0 & \text{otherwise.} \end{cases}$$

$$\mu_3([a, b)) = \begin{cases} 1 & \text{if } a < \frac{1}{\sqrt{2}} < b \\ 0 & \text{otherwise.} \end{cases}$$

Define $\mathbb{A}_j = \{[a_n^{(j)}, b_n^{(j)})\}_{n \in \mathbb{N}}$, ($j = 1 \sim 4$) satisfying

$$a_n^{(1)} \nearrow \frac{1}{2}, b_n^{(1)} \searrow \frac{1}{2}, \quad a_n^{(2)} \nearrow \frac{1}{\sqrt{2}}, b_n^{(2)} \searrow \frac{1}{\sqrt{2}}$$

$$a_n^{(3)} \nearrow \frac{1}{3}, b_n^{(3)} \searrow \frac{1}{3}, \quad a_n^{(4)} \nearrow \frac{1}{\sqrt{3}}, b_n^{(4)} \searrow \frac{1}{\sqrt{3}}.$$

Then, we have

$$\underline{\mu}(\mathbb{A}_j) = \begin{cases} 1 & (j = 1) \\ 1 & (j = 2) \\ 0 & (j = 3) \\ 0 & (j = 4) \end{cases}, \quad \bigcap_{n \in \mathbb{N}} [a_n^{(j)}, b_n^{(j)}) = \begin{cases} \{\frac{1}{2}\} & (j = 1) \\ \emptyset & (j = 2) \\ \{\frac{1}{3}\} & (j = 3) \\ \emptyset & (j = 4) \end{cases}.$$

Thus, elements of $\mathcal{R}(\mathcal{A})$ can fall into four possible categories based on the presence or absence of tangibility (tangible/intangible) and mass (massive/massless).

This website sets only cookies which are necessary for it to function. They are used to enable core functionality such as security, network management and accessibility. These cookies cannot be switched off in our systems. You may disable these by changing your browser settings, but this may affect how the website functions. Please view our privacy policy for further details on how we process your information.

(a) $\underline{\mu}(\mathbb{A}) > 0$.

(b) For any $B \in \mathcal{A}$, we have either $\underline{\mu}(\mathbb{A} \cap B) = \underline{\mu}(\mathbb{A})$ or $\underline{\mu}(\mathbb{A} \cap B) = 0$.

We denote the set of all μ -atoms by $\mathfrak{A}$ and, for each $A \in \mathcal{A}$, $\mathfrak{A}(A)$ denotes the set of all μ -atoms under the restriction to A .

For any $\mathbb{A} \in \mathfrak{A}$ and $\mathbb{B} = \{B_n\}_{n \in \mathbb{N}} \in \mathcal{R}(\mathcal{A})$, either $\underline{\mu}(\mathbb{A} \cap \mathbb{B}) = \underline{\mu}(\mathbb{A})$ or $\underline{\mu}(\mathbb{A} \cap \mathbb{B}) = 0$ is true. (By the definition of μ -atom, $\underline{\mu}(\mathbb{A} \cap B_n) = \underline{\mu}(\mathbb{A})$ for any $n \in \mathbb{N}$ or there exists $n \in \mathbb{N}$ satisfying $\underline{\mu}(\mathbb{A} \cap B_n) = 0$. The latter case implies that $\underline{\mu}(\mathbb{A} \cap \mathbb{B}) = 0$ since the sequence $\{B_n\}_{n \in \mathbb{N}}$ is decreasing.)

Definition 3

Define the following relations on the set of μ -atoms $\mathfrak{A}$ as follows.

$$\begin{aligned} \mathbb{A} \sim \mathbb{B} \text{ if and only if } & \underline{\mu}(\mathbb{A}) = \underline{\mu}(\mathbb{B}) = \underline{\mu}(\mathbb{A} \cap \mathbb{B}) \\ \mathbb{A} \perp \mathbb{B} \text{ if and only if } & \underline{\mu}(\mathbb{A} \cap \mathbb{B}) = 0, \end{aligned}$$

where $\mathbb{A}, \mathbb{B} \in \mathfrak{A}$.

Lemma 1

For the relations $\sim, \perp$ on $\mathfrak{A}$ and $\mathbb{A}, \mathbb{B} \in \mathfrak{A}$, we have

This website sets only cookies which are necessary for it to function. They are used to enable core functionality such as security, network management and accessibility. These cookies cannot be switched off in our systems. You may disable these by changing your browser settings, but this may affect how the website functions. Please view our privacy policy for further details on how we process your information.

(e) $\{\underline{\mu}(A) \mid A \in \mathfrak{A}\}$ attains a maximal value.

Proof. (a) $\sim$ (c) are clear.

For the proof for (d) and (e), it is enough to prove, for any $\varepsilon > 0$

$$\{A \in \mathfrak{A} \mid \underline{\mu}(A) > \varepsilon\} / \sim$$

is a finite set. ($A \sim B$ implies $\underline{\mu}(A) = \underline{\mu}(B)$.)

For the assumption in proof by contradiction, we assume that $\{A \in \mathfrak{A} \mid \underline{\mu}(A) > \varepsilon\} / \sim$ is not finite. Then, for n satisfying $\frac{n\varepsilon}{2} > \mu(X)$, there exists $\{A_j\}_{j=1}^n$ s.t.

$$j, k \leq n, j \neq k \Rightarrow A_j \perp A_k.$$

and

$$\underline{\mu}(A_j) > \varepsilon, \quad \forall j \leq n.$$

Let A_i be expressed as $A_i = \{A_i^{(j)}\}$. Then, for any i, ℓ , we have

This website sets only cookies which are necessary for it to function. They are used to enable core functionality such as security, network management and accessibility. These cookies cannot be switched off in our systems. You may disable these by changing your browser settings, but this may affect how the website functions. Please view our privacy policy for further details on how we process your information.

$$\begin{aligned}
\mu\left(\bigcup_{j=1}^n A_K^{(j)}\right) &= \sum_{j=1}^n \left(\mu\left(A_K^{(j)}\right) - \sum_{\ell=1}^{j-1} \mu\left(A_K^{(j)} \cap A_K^{(\ell)}\right) \right) \\
&\geq \sum_{j=1}^n \left(\varepsilon - \sum_{\ell=1}^{j-1} \frac{\varepsilon}{2n} \right) \\
&\geq n\varepsilon - \frac{(n-1)\varepsilon}{2} \geq \frac{n\varepsilon}{2} > \mu(X).
\end{aligned}$$

This contradicts the monotonicity of μ and thus concludes the proof.

In the process of refining the partition of the space, any μ -atom contained in a set is inherited by one of the resulting subsets. Consequently, the measure of such a subset cannot be smaller than the value of the μ -atom.

3 Concepts on Finitely-Additive Set Functions

First, we introduce the following concept to avoid some unnatural cases.

Definition 4

Let $(X, \mathcal{A}, \mu)$ be a bounded nonnegative finitely-additive set function space.

- (a) The space $(X, \mathcal{A}, \mu)$ is natural if there exist a subfamily $\mathcal{A}' \subset \mathcal{A}$ such that $\mathcal{A} = \mathcal{A}(\mathcal{A}')$ and for every $A \in \mathcal{A}'$, both $\mu(A)$ and $\mu(A^c) > 0$ hold, where $\mathcal{A}(\mathcal{A}')$

This website sets only cookies which are necessary for it to function. They are used to enable core functionality such as security, network management and accessibility. These cookies cannot be switched off in our systems. You may disable these by changing your browser settings, but this may affect how the website functions. Please view our privacy policy for further details on how we process your information.

The following example is a typical non-separable and unnatural set function space.

Example 2

[5] Let X be an infinite set, $\mathcal{A}$ denote the set family of all finite or co-finite sets of X , and define μ as follows.

$$\mu(A) = \begin{cases} 1, & |A^c| < \infty \\ 0, & \text{otherwise.} \end{cases}$$

Then, μ is a bounded, non-negative, and finitely-additive set function on $(X, \mathcal{A})$. This, however, is neither separable nor natural, which is evident from its properties.

The method of extending a family of sets (Boolean algebra) using 0-1-valued additive set functions has been employed in various contexts, including the construction of maximal filters and the σ -additive extension of additive measures [4, 6].

Next, we define $a(A)$, $b(A)$ for each $A \in \mathcal{A}$ as follows.

$$a(A) = \sup \left\{ \mu(\mathbb{A}) \mid \mathbb{A} \in \mathfrak{A}(A) \right\}, \quad b(A) = \mu(A) - a(A).$$

We initially attempted to prove the following condition in full generality or under minimal assumptions. However, to avoid unnecessary complications, we chose to incorporate it as an

This website sets only cookies which are necessary for it to function. They are used to enable core functionality such as security, network management and accessibility. These cookies cannot be switched off in our systems. You may disable these by changing your browser settings, but this may affect how the website functions. Please view our privacy policy for further details on how we process your information.

In particular, when we assume $a(A) = 0$, for any $\varepsilon > 0$, there exists $B \in \mathcal{A}$ such that $B \subset A$ and $|\mu(B) - \mu(A \setminus B)| < \varepsilon$.

Lemma 2

Assume that μ is fundamentally decomposable. Then, for each $A \in \mathcal{A}$ satisfying $b(A) > 0$, there exist $B \subset \mathcal{A}$ s.t.

$$B \subset A, b(B), b(A \setminus B) < \frac{4}{5} b(A).$$

Proof. In the case where $a(A) = 0$, the lemma is clear by Definition 5.

Assume that $a(A) > 0$. By Lemma 1, $\{\underline{\mu}(\mathbb{A}) \mid \mathcal{A} \in \mathfrak{A}(A)\}$ has a maximal value. There exists $\mathbb{A} \in \mathfrak{A}(A)$ such that $\underline{\mu}(\mathbb{A}) = a(A)$. Then, using the assumption $b(A) > 0$, we have that there exists $B_1 \in \mathcal{A}$, satisfying $A \subset B_1$ and

$$a(B_1) = a(A), \quad a(A) \leq \mu(B_1) < a(A) + \frac{1}{5}b(A).$$

Defining $B_2 = A \setminus B_1$, we divide the proof into the following two cases.

Case (a): $a(B_2) < \frac{1}{5}b(A).$

Set $B = B_1$, then $b(B) < \frac{4}{5}b(A)$, and

$$b(A \setminus B) = b(B_2) \leq b(A) - \frac{1}{5}b(A) = \frac{4}{5}b(A)$$

Case (a): Define

$$\alpha = \inf \{ |\mu(D) - \mu(B_2 \setminus D)| \mid D \in \mathcal{A}(B_2) \}.$$

Then, the fundamental decomposability of μ implies $\alpha \leq a(B_2)$, and this implies that there exists $D \in \mathcal{A}(B_2)$ satisfying

$$0 \leq \mu(D) - \mu(B_2 \setminus D) < \frac{1}{5}b(A).$$

Remark that the above inequality also implies that $\mu(B_2 \setminus D) \leq \mu(D)$. Using the above assumption and the fact $B_2 = D \cup (B_2 \setminus D)$, we have $\mu(B_2 \setminus D) \leq \frac{1}{2}\mu(B_2)$. Define $B = B_1 \cup (B_2 \setminus D)$, then $D = A \setminus B$. Therefore,

$$\begin{aligned} b(B) &\leq \frac{1}{5}b(A) + \frac{1}{2}\mu(B_2) \\ &\leq \frac{1}{5}b(A) + \frac{1}{2}b(A) < \frac{4}{5}b(A), \\ b(A \setminus B) &\leq \mu(D) \end{aligned}$$

1

$$\begin{aligned}
 b(B) &\leq \frac{1}{5}b(A) < \frac{4}{5}b(A), \\
 b(A \setminus B) &= b(B_2), \\
 &< \mu(B_2) - \frac{1}{5}b(A) \\
 &\leq b(A) - \frac{1}{5}b(A) = \frac{4}{5}b(A).
 \end{aligned}$$

This concludes the proof.

4 Measurable Closure

We now define the measurable closure of a finitely-additive set function space.

Definition 6

Let $(X, \mathcal{A}, \mu)$ be a nonnegative, bounded, and finitely-additive set function space. Let $\widetilde{X}$ be a set, and φ is a map from $\mathcal{A}$ to $2^{\widetilde{X}}$. Then, $(\widetilde{X}, \varphi)$ is a measurable closure of $(X, \mathcal{A}, \mu)$ if these satisfy the following (a) $\sim$ (e).

- (a) $X \subset \widetilde{X}$, equivalently, there is an injective map $\iota : X \rightarrow \widetilde{X}$.
- (b) $\varphi(\emptyset) = \emptyset$, $\varphi(X) = \widetilde{X}$, and $\iota(A) \subset \varphi(A)$ for each $A \in \mathcal{A}$.
- (c) $A, B \in \mathcal{A}$, $A \cap B = \emptyset \Rightarrow \varphi(A) \cap \varphi(B) = \emptyset$.
- (d) $\forall A, B \in \mathcal{A}$, $\varphi(A \cup B) = \varphi(A) \cup \varphi(B)$.

This website sets only cookies which are necessary for it to function. They are used to enable core functionality such as security, network management and accessibility. These cookies cannot be switched off in our systems. You may disable these by changing your browser settings, but this may affect how the website functions. Please view our privacy policy for further details on how we process your information.

Taking the rational numbers as the underlying space, a finitely-additive set function that assigns length to the algebra generated by finite unions of intervals cannot be extended to a countably additive measure as it is. However, if the space is extended to the real numbers, such a countably additive extension exists. The concept of measurable closure aims to generalize this idea to arbitrary finitely-additive measure spaces.

The following lemma presents a property that can be derived from existing results using standard techniques. Thus, we provide only a guideline for the proof.

Lemma 3

A measurable closure $(\widetilde{X}, \varphi)$ of $(X, \mathcal{A}, \mu)$ satisfies the following (a) $\sim$ (c).

(a) $\widetilde{\mathcal{A}} = \{\varphi(A) \mid A \in \mathcal{A}\}$ is an algebra.

(b) $\widetilde{\mu}(\widetilde{A}) = \mu(A)$ is σ -additive on $\widetilde{\mathcal{A}}$.

(c) For a subset $\widetilde{A} \subset \widetilde{X}$, define

$$\widetilde{\mu}^\sigma(\widetilde{A}) = \inf \left\{ \sum_{n=1}^{\infty} \widetilde{\mu}(\widetilde{A}_n) \mid \{\widetilde{A}_n\}_{n \in \mathbb{N}} \subset \widetilde{\mathcal{A}}, \widetilde{A} \subset \bigcup_{n=1}^{\infty} \widetilde{A}_n \right\},$$

then,

$$\widetilde{\mathcal{A}}^\sigma = \left\{ \widetilde{A} \mid \widetilde{\mu}(\widetilde{B}) = \mu(\widetilde{B} \cap \widetilde{A}) + \mu(\widetilde{B} \cap \widetilde{A}^c), \forall \widetilde{B} \subset \widetilde{X} \right\}$$

This website sets only cookies which are necessary for it to function. They are used to enable core functionality such as security, network management and accessibility. These cookies cannot be switched off in our systems. You may disable these by changing your browser settings, but this may affect how the website functions. Please view our privacy policy for further details on how we process your information.

follows from Corollary 2.3 in [1].

We also denote the restriction of $\tilde{\mu}^\sigma$ to $\tilde{\mathcal{B}} = \tilde{\mathcal{A}}^\sigma$ by $\tilde{\mu}$. The σ -additive measure space $(\tilde{X}, \tilde{\mathcal{B}}, \tilde{\mu})$ is also called the measurable closure.

Example 3

Let $(X, \mathcal{A}, \mu)$ be the same set function space with Example 1.

$(X = [0, 1] \cap \mathbb{Q})$. Define

$$\begin{aligned}\tilde{X} &= [0, 1] \\ \varphi([a, b] \cap \mathbb{Q}) &= [a, b], \quad \forall a, b \in X.\end{aligned}$$

Then, we can extend the map φ to $\mathcal{A}$, and $(\tilde{X}, \varphi)$ is a measurable closure. The extended measure $\tilde{\mu}$ is given by

$$\tilde{\mu} = \lambda + \delta_{\frac{1}{2}} + \delta_{\frac{1}{\sqrt{2}}},$$

where λ is the Lebesgue measure on $[0, 1]$, and $\delta_{\frac{1}{2}}$ and $\delta_{\frac{1}{\sqrt{2}}}$ denote the Dirac measures at $\frac{1}{2}$ and $\frac{1}{\sqrt{2}}$, respectively.

Theorem 1

This website sets only cookies which are necessary for it to function. They are used to enable core functionality such as security, network management and accessibility. These cookies cannot be switched off in our systems. You may disable these by changing your browser settings, but this may affect how the website functions. Please view our privacy policy for further details on how we process your information.

(a) Recursive partition, which will be used to construct the extension space.

(b) Definition of the extension space $\widetilde{X}$.

(c) Definition of the mapping φ .

(d) Extension of the measure.

(e) Final verification of measure consistency and σ -additivity.

(a) We may assume that $A_1 = X$. We will define a sequence of finite partitions

$\left\{ \mathbb{D}_m = \left\{ D_k^{(m)} \right\}_{k=1}^{N_m} \right\}_{m \in \mathbb{N}}$ ensuring that each $\mathbb{D}_m$ is a refinement of $\mathbb{D}_{m-1}$. The construction proceeds as follows.

Let $\mathbb{D}_1 = \{A_1 (= X)\}$, and define $\alpha = \frac{4}{5}$. Set $M = b(A_1)$. Then, it is clear that for any $D \in \mathbb{D}_1$, $b(D) \leq M\alpha^0 = M$ holds. Assume that we have obtained $\mathbb{D}_1 \sim \mathbb{D}_{m-1}$, satisfying the condition that for any $k \leq m-1$ and any $D \in \mathbb{D}_k$, $b(D) \leq M\alpha^{k-1}$. We then set $\mathbb{D}' = \mathbb{D}_{m-1}$. Clearly, for any $D \in \mathbb{D}'$, we have $b(D) \leq M\alpha^{m-2}$.

In the case where $A_m \notin \mathcal{A}(\mathbb{D}')$, where $\mathcal{A}(\mathbb{D}')$ is the smallest algebra containing $\mathbb{D}'$, consider the refinement

$$\{D \cap A_m, D \cap A_m^c \mid D \in \mathbb{D}'\},$$

and this refinement is also denoted by $\mathbb{D}'$.

Thus, by induction, we obtain a sequence of partitions $\{\mathbb{D}_m\}_{m \in \mathbb{N}}$ satisfying, for each $m \in \mathbb{N}$, the condition

$$b(D) \leq M\alpha^{m-1}$$

for all $D \in \mathbb{D}_m$.

(b) Next, we consider the following family of decreasing sequences of sets.

$$\widetilde{\mathcal{X}} = \left\{ \mathcal{S} = \left\{ D_{k_m}^{(m)} \right\}_{m \in \mathbb{N}} \mid \begin{array}{l} \{k_m\}_{m \in \mathbb{N}} \in \prod_{m \in \mathbb{N}} \{1, 2, \dots, N_m\}, \\ D_{k_{m+1}}^{(m+1)} \subset D_{k_m}^{(m)} \forall m \in \mathbb{N} \end{array} \right\} \subset \mathcal{R}.$$

Next, we define $\iota; \mathcal{X} \rightarrow \widetilde{\mathcal{X}}$ as follows. Fix arbitrary $x \in \mathcal{X}$ and $m \in \mathbb{N}$. Then, there uniquely exists $j(m, x) \leq N_m$ such that $x \in D_{j(m, x)}^{(m)}$. Define

$\iota(x) = \left\{ D_{j(m, x)}^{(m)} \right\}_{m \in \mathbb{N}} \in \widetilde{\mathcal{X}}$. Then, as we defined in Definition 4, the separability implies that ι is injective.

(c) From the construction of the partition $\{\mathbb{D}_m\}_{m \in \mathbb{N}}$, for each $A \in \mathcal{A}$, there exist $m \in \mathbb{N}$ and $\mathbb{D}' \subset \mathbb{D}_m$ such that $A = \cup_{\mathbb{D}'}$. We can define

Then, we obtain that $\widetilde{\mathcal{A}}$ is an algebra.

(d) Using μ -atoms, we define

$$b'(A) = \mu(A) - \sum_{\mathbb{A} \in \mathfrak{A}, \mathbb{A} \subset A} \mu(\mathbb{A}),$$

where $\mathbb{A} = \{A_n\}_{n \in \mathbb{N}} \subset A$ means that $A_n \subset A$ for sufficiently large n . Remark that we have $b'(A) \leq b(A)$ for any $A \in \mathcal{A}$.

Using the fact that two different atoms $\mathbb{A}_1, \mathbb{A}_2 \in \widetilde{\mathcal{X}}$ satisfy $\mathbb{A}_1 \perp \mathbb{A}_2$ and Lemma 1, we have

$$\mu(A) = b'(A) + \sum_{\mathbb{A} \in \widetilde{\mathcal{X}}, \mathbb{A} \subset A} \mu(\mathbb{A}).$$

Set $I = [0, b'(X)]$ and $\mathfrak{A}' = \{\mathbb{A} \in \widetilde{\mathcal{X}} \mid \mathbb{A} \in \mathfrak{A}\}$. Let λ be the Lebesgue measure on I , and define a σ -additive measure ν on $\mathfrak{A}'$ as follows.

$$\nu(U) = \sum_{\mathbb{A} \in U} \mu(\mathbb{A}), \quad U \subset \mathfrak{A}'.$$

.

(m)

This website sets only cookies which are necessary for it to function. They are used to enable core functionality such as security, network management and accessibility. These cookies cannot be switched off in our systems. You may disable these by changing your browser settings, but this may affect how the website functions. Please view our privacy policy for further details on how we process your information.

Thus, we defined a mapping ψ from $\mathcal{A}'$ to $\mathcal{B}(I) \cup 2^{\mathcal{A}'}$. By the above definitions, for each $A \in \mathcal{A}$, we have

$$\mu(A) = \tilde{\mu}(\varphi(A)) = (\lambda + \nu)(\psi(\varphi(A))).$$

This completes the construction of an additive measure $\tilde{\mu}$ on the extension space $\widetilde{X}$.

(e) Suppose that $\{A_n\}_{n \in \mathbb{N}} \subset \mathcal{A}$ and $A \in \mathcal{A} (\{\varphi(A_n)\}_{n \in \mathbb{N}} \subset \widetilde{\mathcal{A}}, \varphi(A) \in \widetilde{\mathcal{A}})$ satisfy

$$\varphi(A) = \bigcup_{n=1}^{\infty} \varphi(A_n),$$

we have

$$\psi(\varphi(A)) = \bigcup_{n=1}^{\infty} \psi(\varphi(A_n)).$$

This equality is key to the proof, and its validity can be shown by assigning to each $A \in \widetilde{X}$ a Cauchy sequence in I . Although the sum on the right-hand side is not necessarily disjoint, the measure of each pairwise intersection is zero, ensuring countable additivity. We

$$\tilde{\mu}(\varphi(A)) = \sum_{n=1}^{\infty} \tilde{\mu}(A_n).$$

Thus, we have the σ -additivity of $\tilde{\mu}$ and this concludes the proof.

5 Conclusion

In this paper, we introduced a novel approach for extending finitely-additive set functions to σ -additive measures through the construction of a measurable closure. Unlike classical methods based on Stone spaces, our framework emphasizes intuitive and constructive techniques that avoid reliance on non-constructive tools such as Zorn's lemma.

By systematically refining partitions and utilizing the structure of μ -atoms, we demonstrated how to embed a given finitely-additive measure space into a larger space in which σ -additivity naturally emerges. The construction of a measurable closure allows for a seamless transition from algebraic to topological and measure-theoretic perspectives.

Furthermore, we showed that under assumptions such as separability and fundamental decomposability, the existence of a measurable closure is guaranteed. This provides a concrete path to understand the limitations and possibilities of extending additive measures, shedding new light on the interplay between algebraic structure and measure-theoretic completeness.

We hope that this framework serves not only as a tool for analysis but also as a foundation for

Möbius transforms given as additive set functions.

References

1. Osborne, M.J., Rubinstein, A.: A Course in Game Theory. MIT Press, Cambridge (1994)

[MATH](#) [Google Scholar](#)

2. Aumann, R.J., Shapley, L.S.: Values of Non-Atomic Games. Princeton University Press, Princeton (1974)

[MATH](#) [Google Scholar](#)

3. Tao, T.: An Introduction to Measure Theory, American Mathematical Society, Graduate Studies in Mathematics 126

[Google Scholar](#)

4. Denneberg, D.: Representation of the Choquet integral with the σ -additive Möbius transform. Fuzzy Sets Syst. **92**, 139–156 (1997)

[Article](#) [MathSciNet](#) [Google Scholar](#)

5. Bhaskara Rao, K., Bhaskara Rao, M.: Theory of Charges: A Study of Finitely Additive Measures. Academic Press, London (1983)

This website sets only cookies which are necessary for it to function. They are used to enable core functionality such as security, network management and accessibility. These cookies cannot be switched off in our systems. You may disable these by changing your browser settings, but this may affect how the website functions. Please view our privacy policy for further details on how we process your information.

7. Denneberg, D.: Non-additive measure and integral. Theory and Decision Library Series B, vol. 27, Kluwer Academic, Dordrecht (1994)

[Google Scholar](#)

8. Fukuda, R., Honda, A., Okazaki, Y.: An axiomatic definition of non-discrete Möbius transform. In: Lecture Notes in Computer Science (14986), pp. 19–30 (2024)

[Google Scholar](#)

Acknowledgements

The authors would like to express their sincere gratitude to Professor J. Kawabe and Professor K. Fujimoto for introducing the book [5], which provided valuable insights into finitely additive measures.

We are also grateful to the three anonymous referees for their constructive comments and suggestions, which significantly improved the quality of this paper.

Author information

Authors and Affiliations

Oita University, 700 Dan-noharu, Oita, 870-1192, Japan

This website sets only cookies which are necessary for it to function. They are used to enable core functionality such as security, network management and accessibility. These cookies cannot be switched off in our systems. You may disable these by changing your browser settings, but this may affect how the website functions. Please view our privacy policy for further details on how we process your information.

Corresponding author

Correspondence to [Ryoji Fukuda](#).

Editor information

Editors and Affiliations

Umeå University, Umeå, Sweden

Vicenç Torra

Tamagawa University, Tokyo, Japan

Yasuo Narukawa

Universitat Rovira i Virgili, Tarragona, Spain

Josep Domingo-Ferrer

Rights and permissions

[Reprints and permissions](#)

Copyright information

© 2026 The Author(s), under exclusive license to Springer Nature Switzerland AG

About this paper

This website sets only cookies which are necessary for it to function. They are used to enable core functionality such as security, network management and accessibility. These cookies cannot be switched off in our systems. You may disable these by changing your browser settings, but this may affect how the website functions. Please view our privacy policy for further details on how we process your information.

[Download book PDF](#)[Download book EPUB](#)[.RIS↓](#) [.ENW↓](#) [.BIB↓](#)

DOI
https://doi-org.proxy.lib.sfu.ca/10.1007/978-3-032-00891-6_1

Published
07 August 2025

Publisher Name
Springer, Cham

Print ISBN
978-3-032-00890-9

Online ISBN
978-3-032-00891-6

eBook Packages
[Computer Science](#)
[Computer Science \(R0\)](#)
[Springer Nature](#)
[Proceedings Computer Science](#)

Share this paper

Anyone you share the following link with will be able to read this content:

[Get shareable link](#)

Provided by the Springer Nature SharedIt content-sharing initiative

Keywords

[finitely-additive measure](#) [\$\sigma\$ -additive expansion](#)[Stone space](#)

This website sets only cookies which are necessary for it to function. They are used to enable core functionality such as security, network management and accessibility. These cookies cannot be switched off in our systems. You may disable these by changing your browser settings, but this may affect how the website functions. Please view our privacy policy for further details on how we process your information.

Download book PDF

Download book EPUB

This website sets only cookies which are necessary for it to function. They are used to enable core functionality such as security, network management and accessibility. These cookies cannot be switched off in our systems. You may disable these by changing your browser settings, but this may affect how the website functions. Please view our privacy policy for further details on how we process your information.